Surgical Copilot

Live Inference Documentation

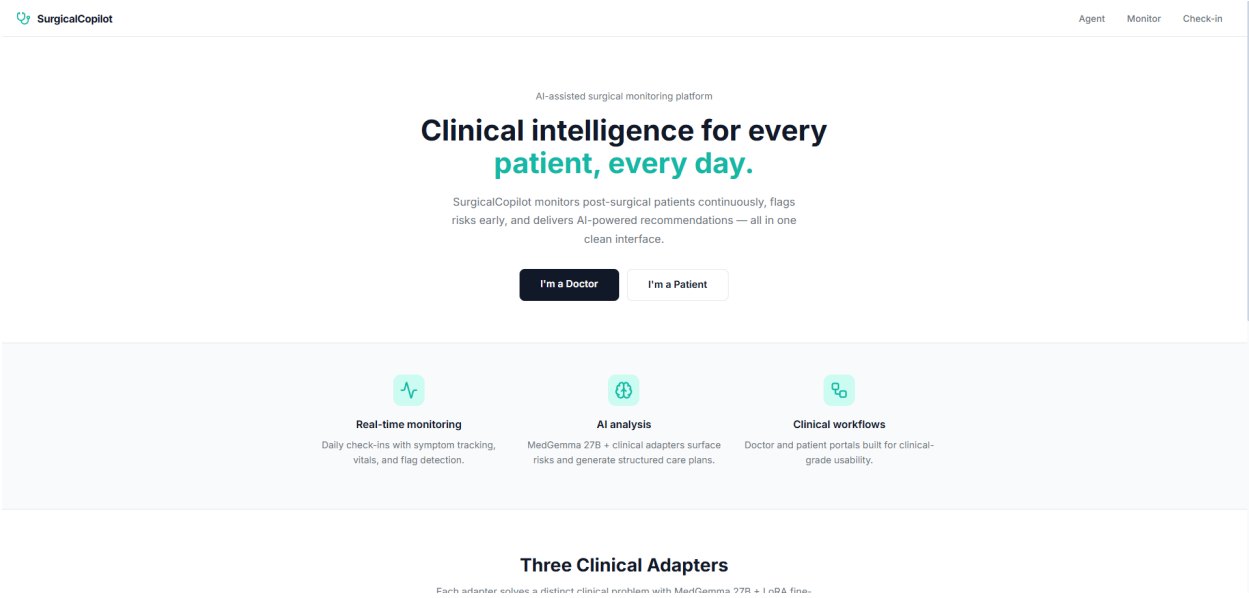
MedGemma-27B + LoRA Adapters | FHIR R4 Integration | Wearable Monitoring

This document provides visual evidence of Surgical Copilot's live inference capabilities. All screenshots show real AI processing using MedGemma-27B with three specialized LoRA adapters for inpatient triage, post-discharge monitoring, and oncology surveillance.

Feature	Technology
Core Reasoning	MedGemma-27B with LoRA adapters
Enrichment Layer	MedGemma-4B for SBAR, patient messages, vision
Voice Input	MedASR medical speech recognition
EHR Integration	FHIR R4 (Epic/Cerner simulation)
Safety	Rule Sentinel + Human-in-the-Loop gates
Wearable Support	Apple Watch, Fitbit, Garmin (simulated)

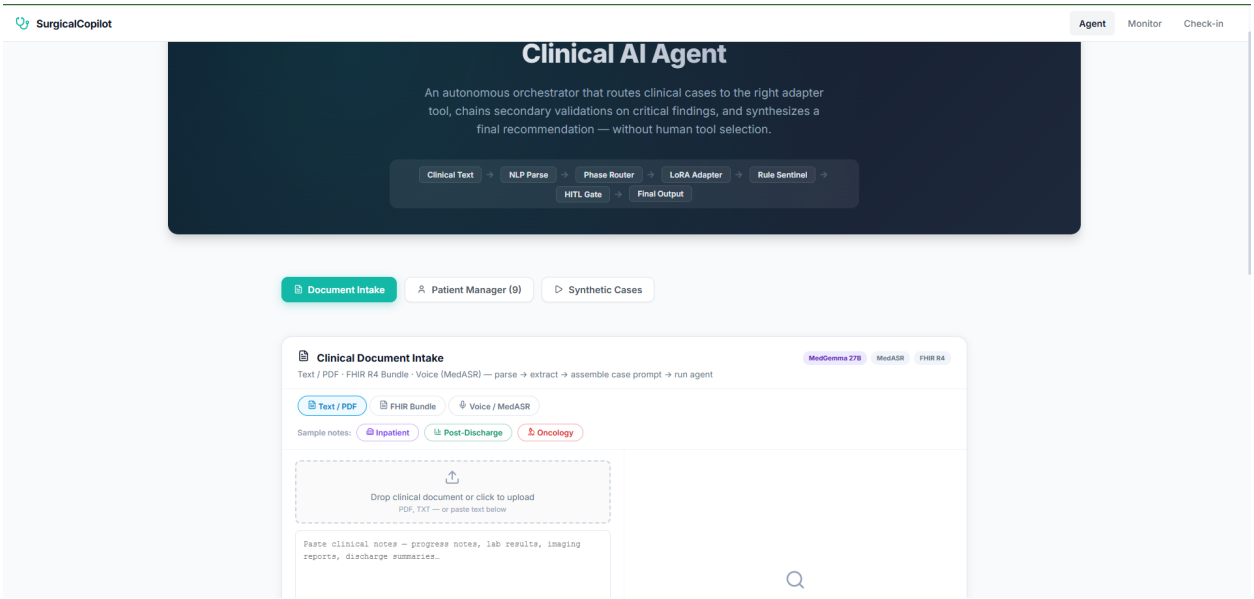
1. Landing Page

Clean, professional landing page introducing Surgical Copilot. Features role-based entry points ("I'm a Doctor" / "I'm a Patient") and highlights three core capabilities: Real-time monitoring, AI analysis powered by MedGemma-27B, and clinical workflows.



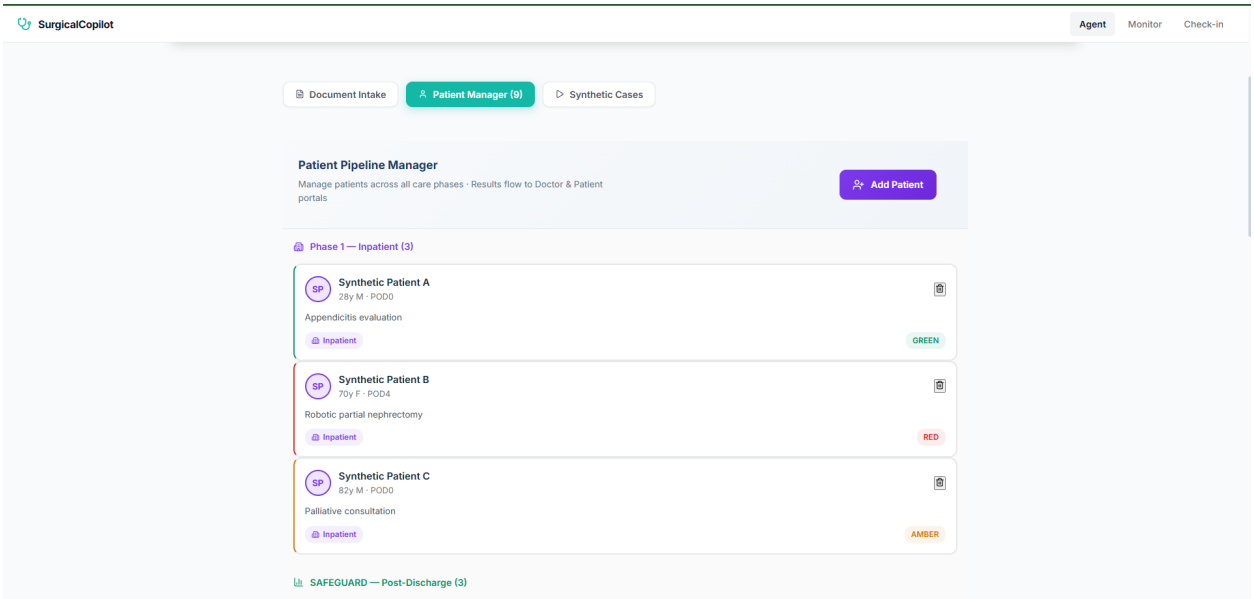
2. Clinical AI Agent - Overview

The Clinical AI Agent page showing the autonomous orchestration pipeline: Clinical Text → NLP Parse → Phase Router → LoRA Adapter → Rule Sentinel → HITL Gate → Final Output. Supports four input methods: Text/PDF, FHIR Bundle, and Voice (MedASR).



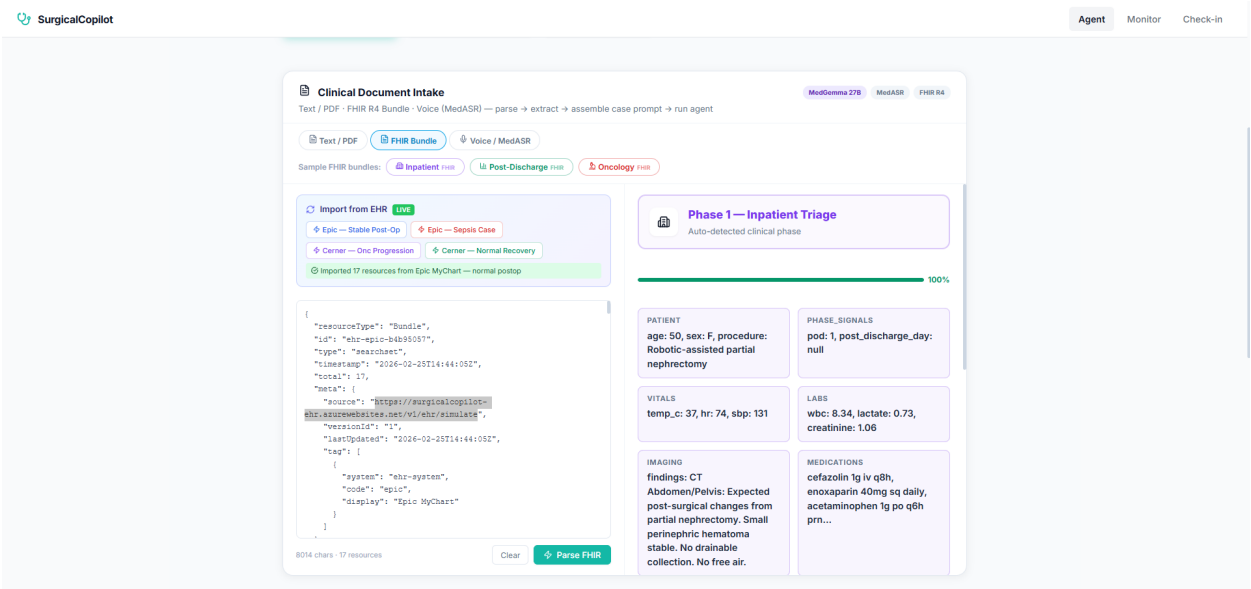
3. Patient Pipeline Manager

Centralized patient management across all care phases. Shows patients grouped by phase (Phase 1 Inpatient, SAFEGUARD Post-Discharge) with color-coded risk badges (GREEN, AMBER, RED). Each card displays procedure, post-op day, and current risk status.



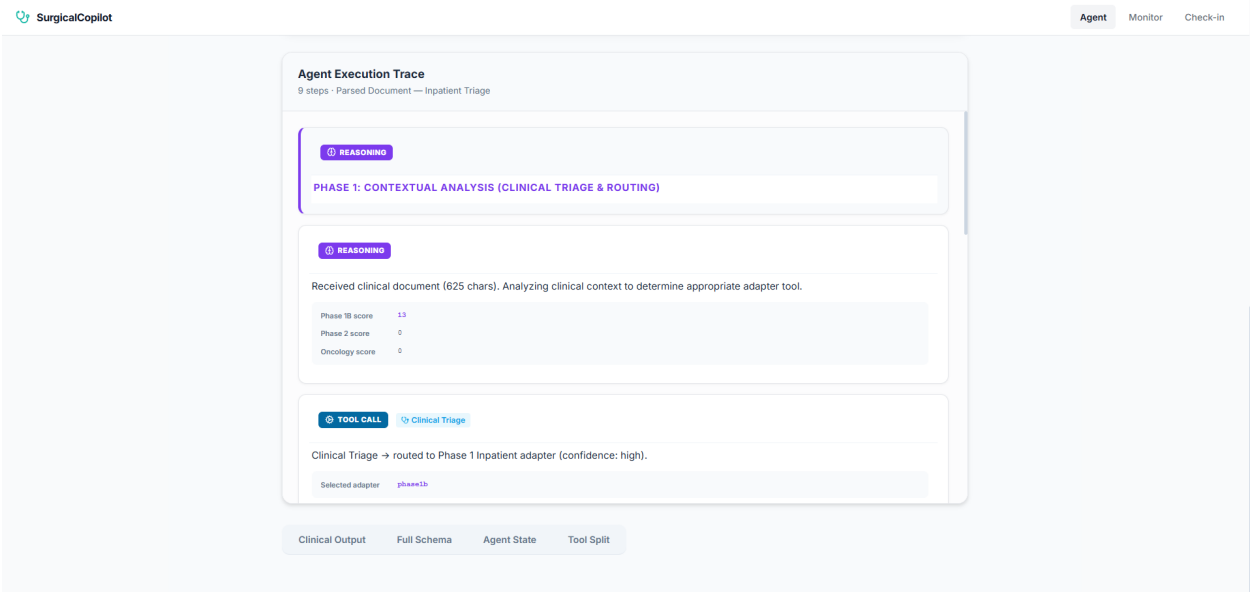
4. FHIR R4 Integration

Live EHR integration via FHIR R4. Shows "Import from EHR" with Epic and Cerner simulated endpoints. Successfully imported 17 FHIR resources including Patient, Observations, DiagnosticReports, and MedicationRequests. Auto-detects clinical phase (Phase 1 Inpatient Triage).



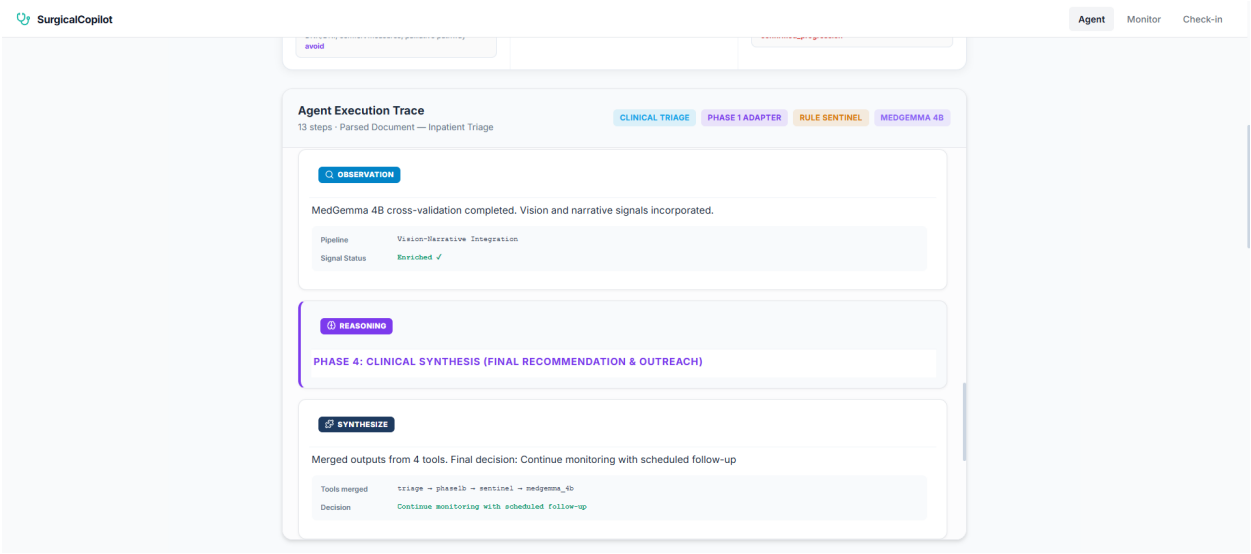
5. Agent Execution Trace - Phase Routing

Real-time agent reasoning trace showing Phase 1 contextual analysis. The router scores each adapter (Phase 1B: 13, Phase 2: 0, Oncology: 0) and routes to the correct LoRA adapter with high confidence. Demonstrates transparent AI decision-making.



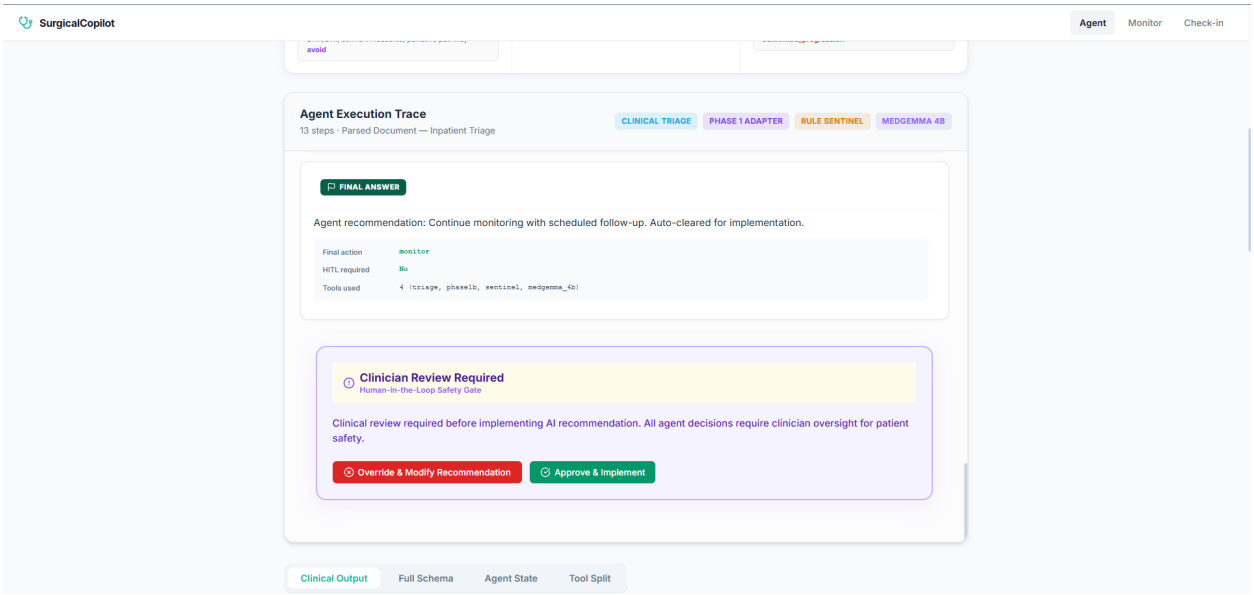
6. Agent Trace - Multi-Tool Synthesis

Shows MedGemma-4B cross-validation and final synthesis. Four tools merged (triage → phase1b → sentinel → medgemma_4b). Final decision: "Continue monitoring with scheduled follow-up." The pipeline shows Vision-Narrative Integration enrichment.



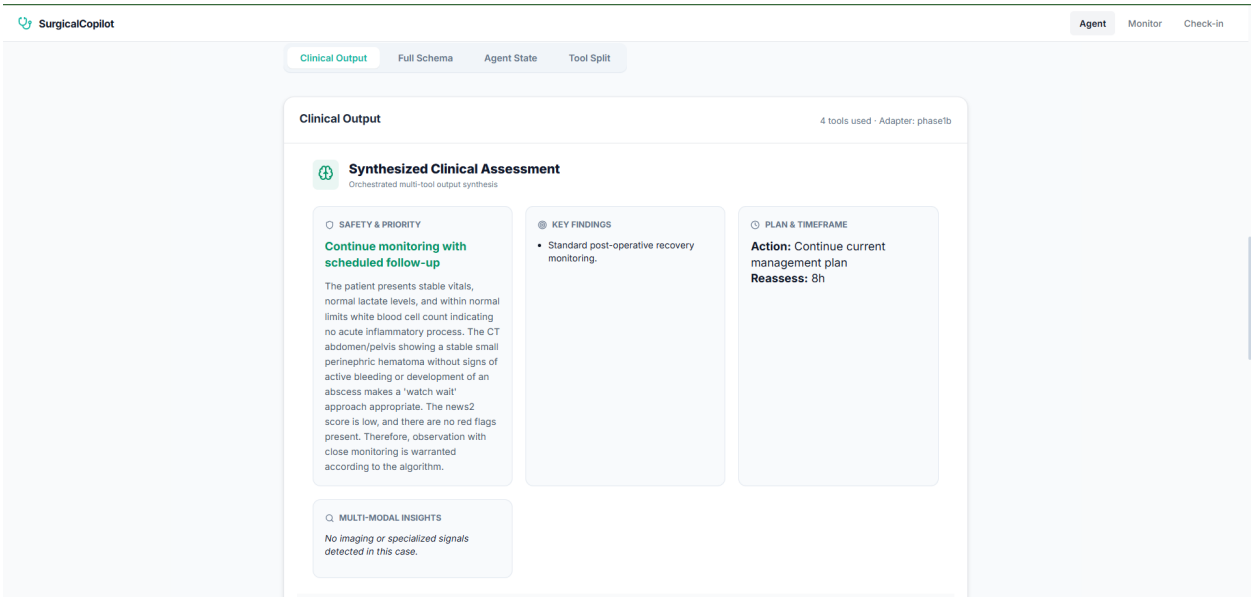
7. Human-in-the-Loop Safety Gate

Critical safety feature: "Clinician Review Required" gate. Shows Final Answer with action="monitor" and HITL required status. Two options: "Override & Modify Recommendation" (red) or "Approve & Implement" (green). All AI decisions require clinician oversight.



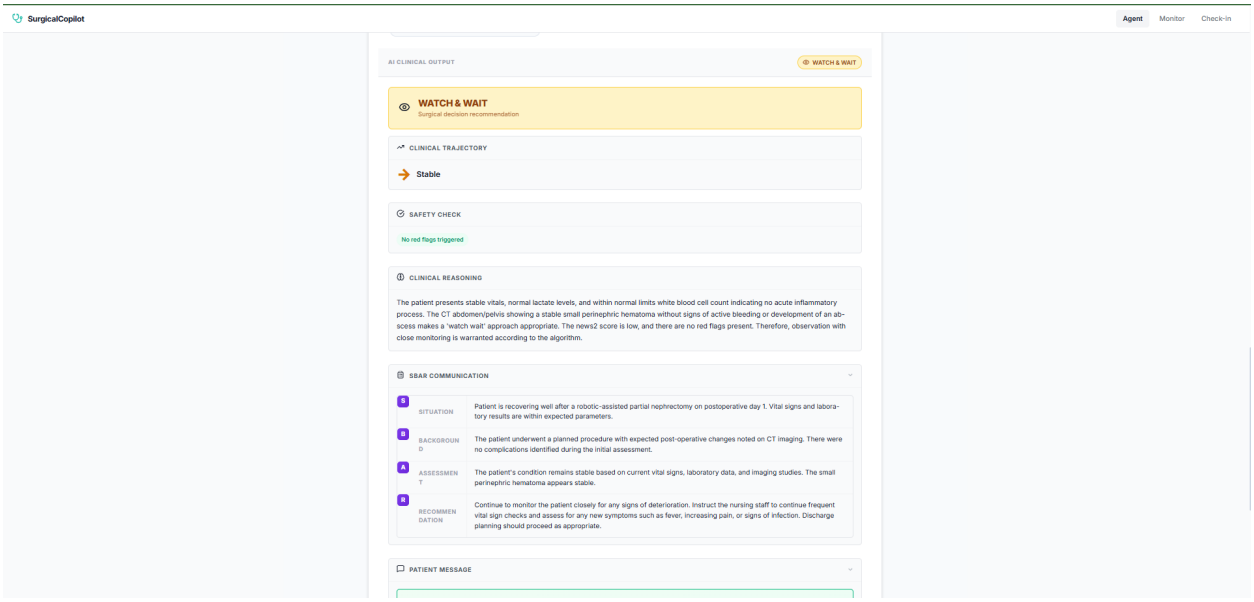
8. Synthesized Clinical Assessment

Structured clinical output with three sections: Safety & Priority (continue monitoring), Key Findings (standard post-op recovery), and Plan & Timeframe (reassess in 8h). Multi-modal insights section shows no specialized signals detected.



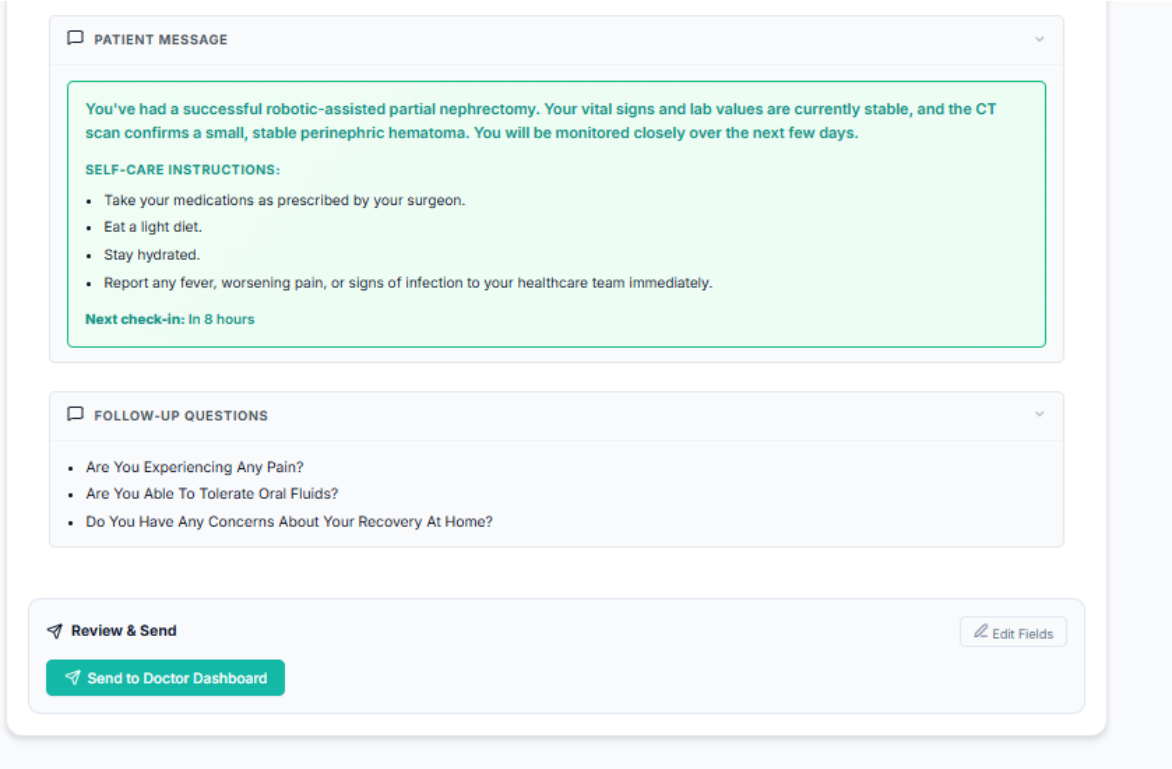
9. AI Clinical Output - Full View

Complete AI output showing: WATCH & WAIT decision (yellow banner), Clinical Trajectory: Stable, Safety Check: No red flags triggered, full Clinical Reasoning, SBAR Communication (Situation/Background/Assessment/Recommendation), and Patient Message section.



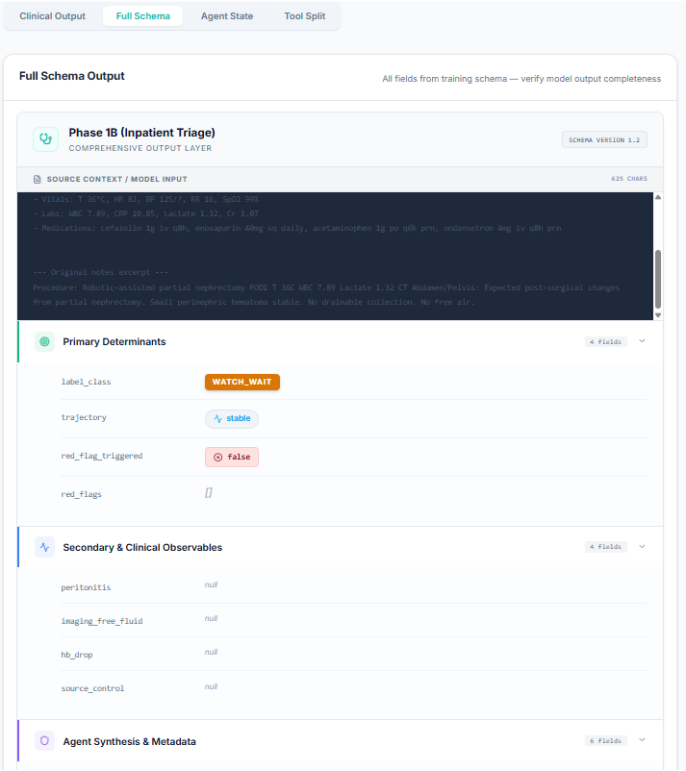
10. Patient-Facing Message

AI-generated patient communication with plain-language summary, Self-Care Instructions (medications, diet, hydration, warning signs), Next check-in countdown (8 hours), and Follow-up Questions. Includes "Send to Doctor Dashboard" action.



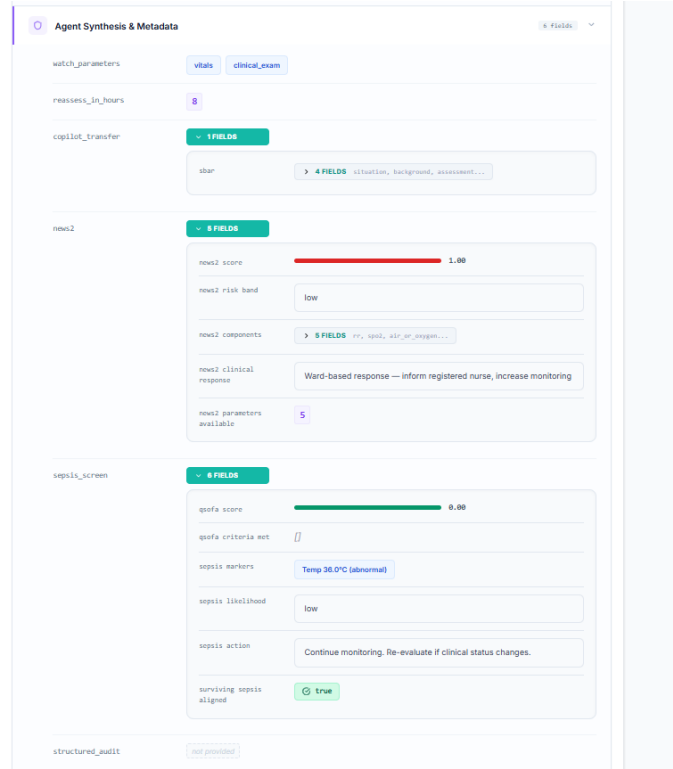
11. Full Schema Output - Primary Determinants

Complete model output schema showing Phase 1B (Inpatient Triage) comprehensive output layer. Primary Determinants: label_class=WATCH_WAIT, trajectory=stable, red_flag_triggered=false. Secondary observables: peritonitis, imaging_free_fluid, hb_drop, source_control all null.



12. Full Schema - Scoring & Sepsis Screen

Detailed enrichment output: NEWS2 score=1.00 (low risk band), clinical response="ward-based, increase monitoring". Sepsis screen: qSOFA=0.00, sepsis likelihood=low, sepsis action="continue monitoring". Surviving Sepsis Campaign aligned=true.



13. Raw JSON Output

Raw model output payload showing structured JSON with label_class, trajectory, red_flag_triggered, SBAR fields, and complete clinical reasoning. Demonstrates deterministic, parseable outputs - no free text hallucination risk.

HIDE RAW OUTPUT PAYLOAD

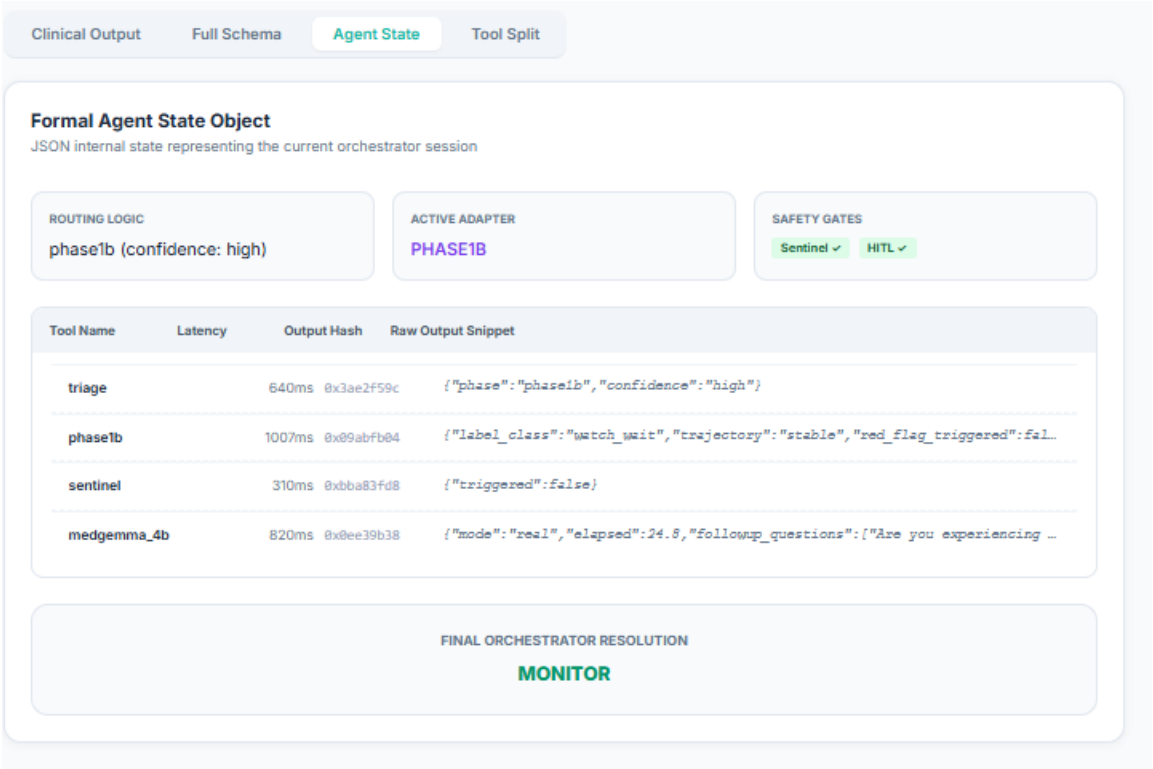
raw_output_trace.json

Copy


```
{
  "label_class": "watch_wait",
  "trajectory": "stable",
  "red_flag_triggered": false,
  "red_flags": [],
  "peritonitis": null,
  "imaging_free_fluid": null,
  "hb_drop": null,
  "source_control": null,
  "watch_parameters": [
    "vitals",
    "clinical_exam"
  ],
  "reassess_in_hours": 8,
  "copilot_transfer": {
    "sbar": {
      "situation": "Patient is recovering well after a robotic-assisted partial nephrectomy on postoperative day 1. Vital signs and l
      "background": "The patient underwent a planned procedure with expected post-operative changes noted on CT imaging. There were n
      "assessment": "The patient's condition remains stable based on current vital signs, laboratory data, and imaging studies. The s
      "recommendation": "Continue to monitor the patient closely for any signs of deterioration. Instruct the nursing staff to contin
```

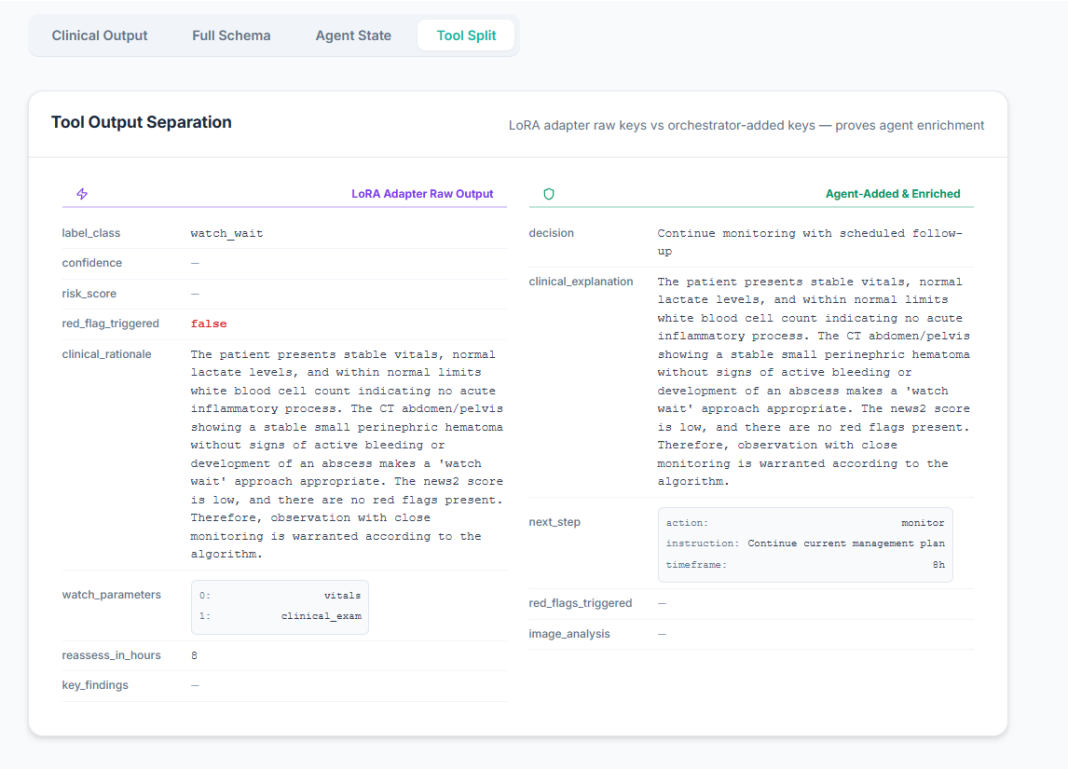
14. Agent State - Pipeline Breakdown

Formal agent state showing: Routing Logic (phase1b, confidence: high), Active Adapter (PHASE1B), Safety Gates (Sentinel ✓, HITL ✓). Tool latencies: triage 640ms, phase1b 1007ms, sentinel 310ms, medgemma_4b 820ms. Final resolution: MONITOR.



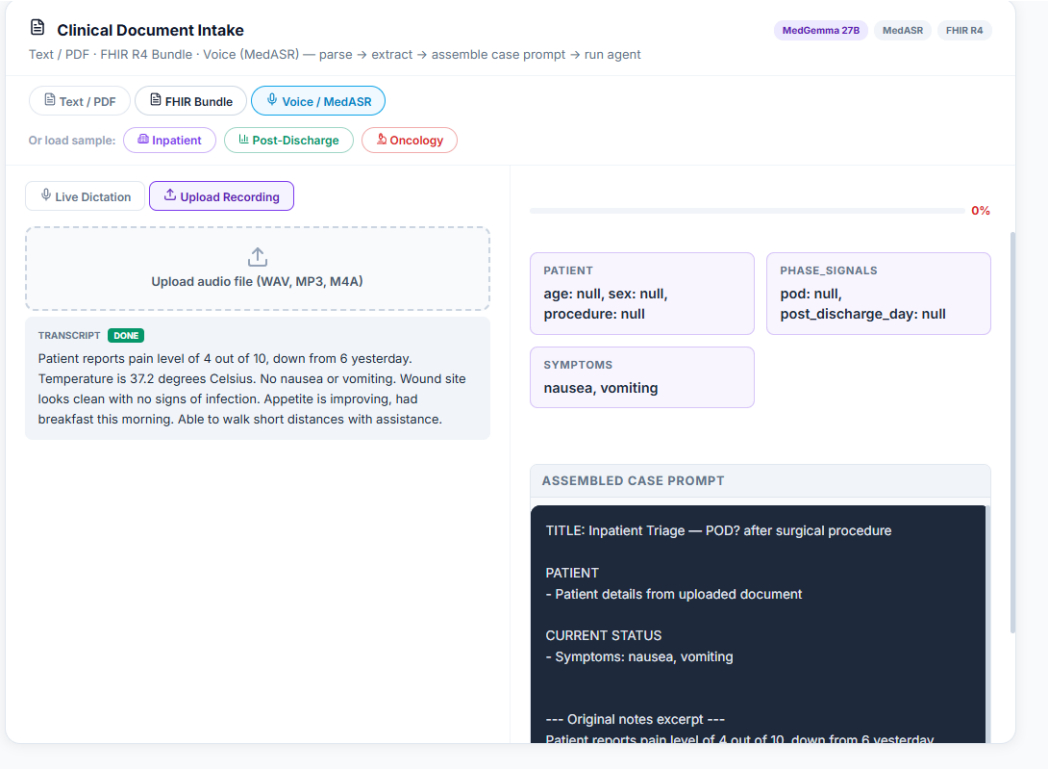
15. Tool Split - Model vs Enrichment

Critical transparency view separating LoRA Adapter Raw Output (label_class, clinical_rationale) from Agent-Added & Enriched content (decision, clinical_explanation, next_step). Proves deterministic enrichment layer adds clinical scoring without model hallucination.



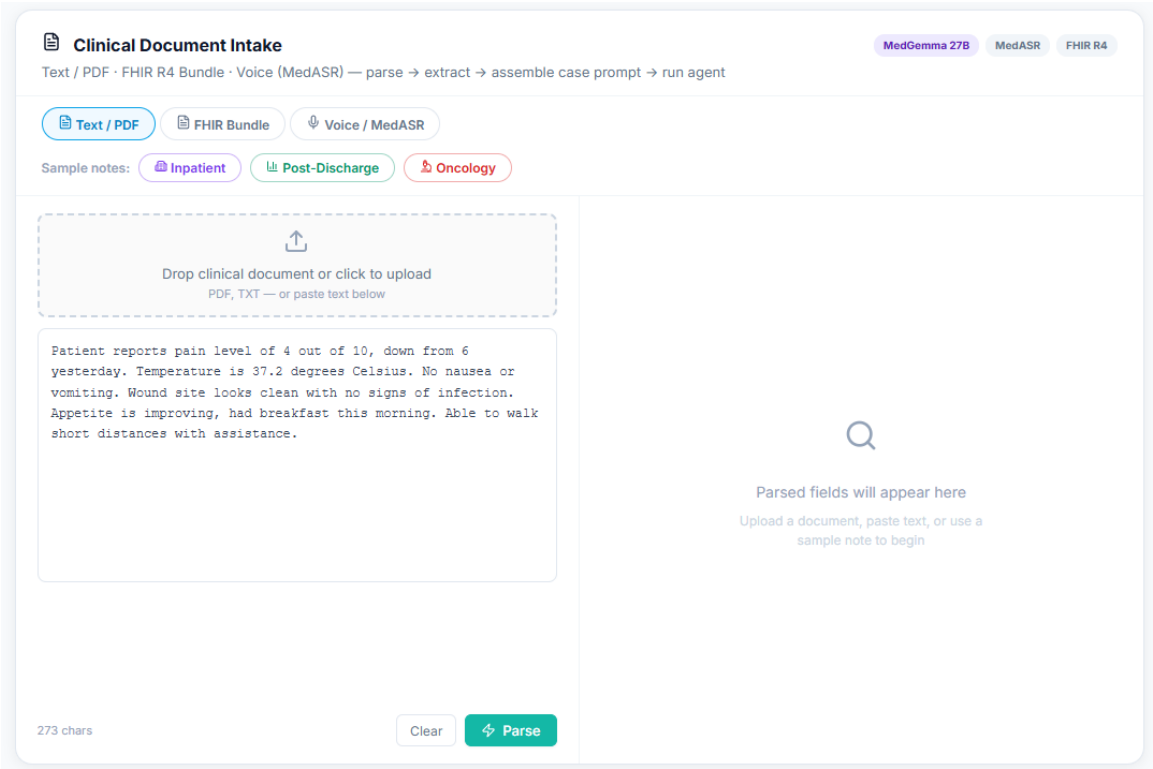
16. Voice Input - MedASR Integration

Voice/MedASR input mode showing medical speech recognition. Transcript displays dictated clinical notes about pain levels, temperature, wound status, appetite, and mobility. Demonstrates hands-free clinical documentation capability.



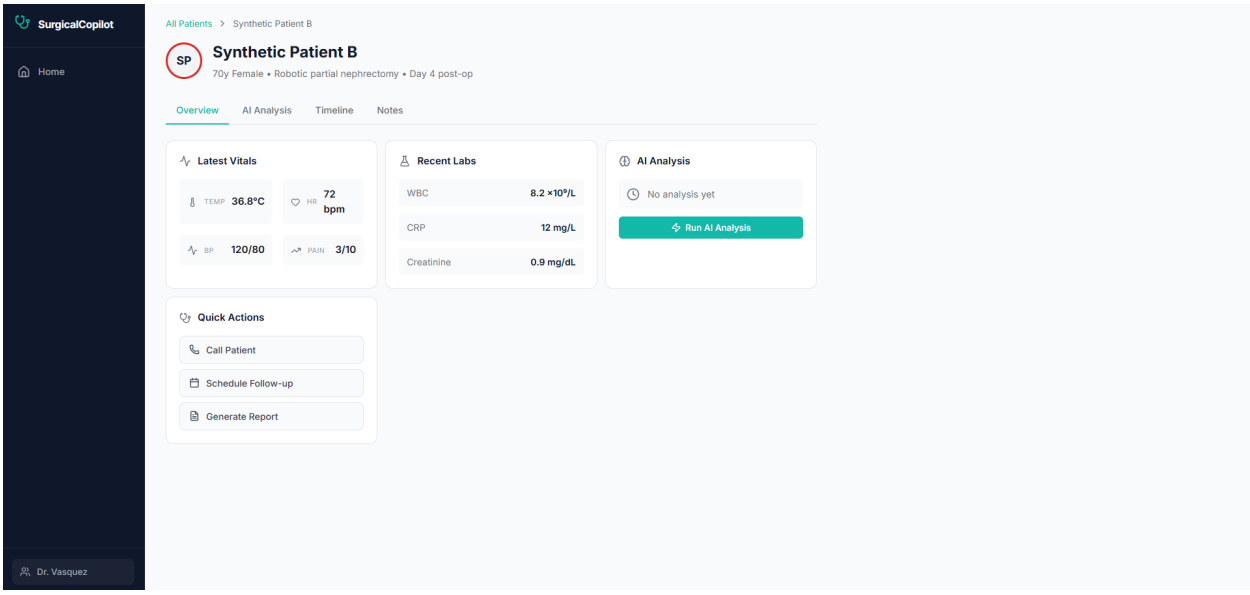
17. Text/PDF Input Mode

Text/PDF document intake showing pasted clinical notes ready for parsing. Sample notes include Inpatient, Post-Discharge, and Oncology examples. 273 characters ready to parse with Clear and Parse buttons.



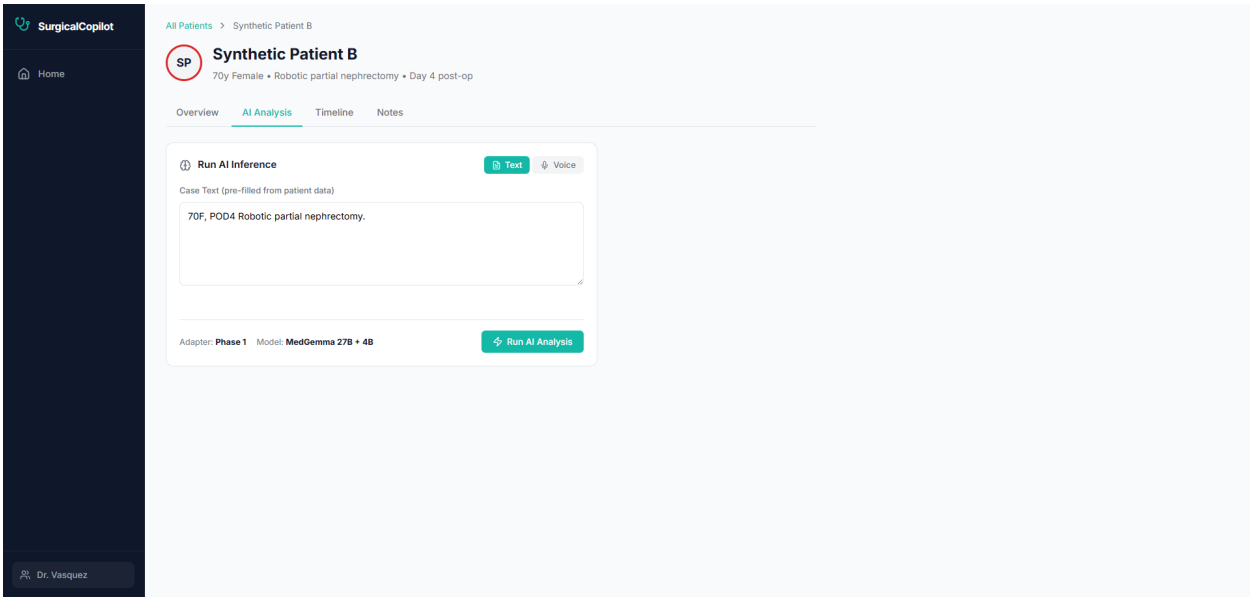
18. Doctor Portal - Patient Overview

Clinical dashboard for Synthetic Patient B (70y Female, robotic nephrectomy, POD4). Shows Latest Vitals (temp 36.8°C, HR 72, BP 120/80, pain 3/10), Recent Labs (WBC, CRP, creatinine), and Quick Actions (Call Patient, Schedule Follow-up, Generate Report).



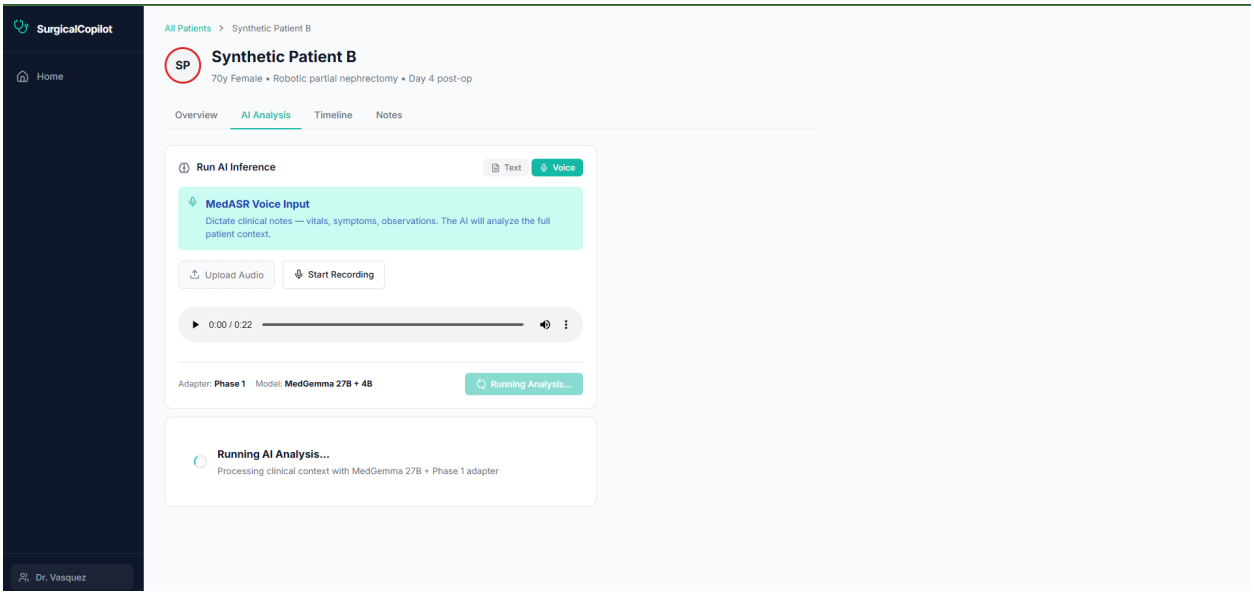
19. Doctor Portal - AI Analysis Tab

Run AI Inference section in Doctor Portal. Pre-filled case text from patient data, Text/Voice toggle, Adapter selection (Phase 1), Model indicator (MedGemma 27B + 4B). One-click "Run AI Analysis" integration.



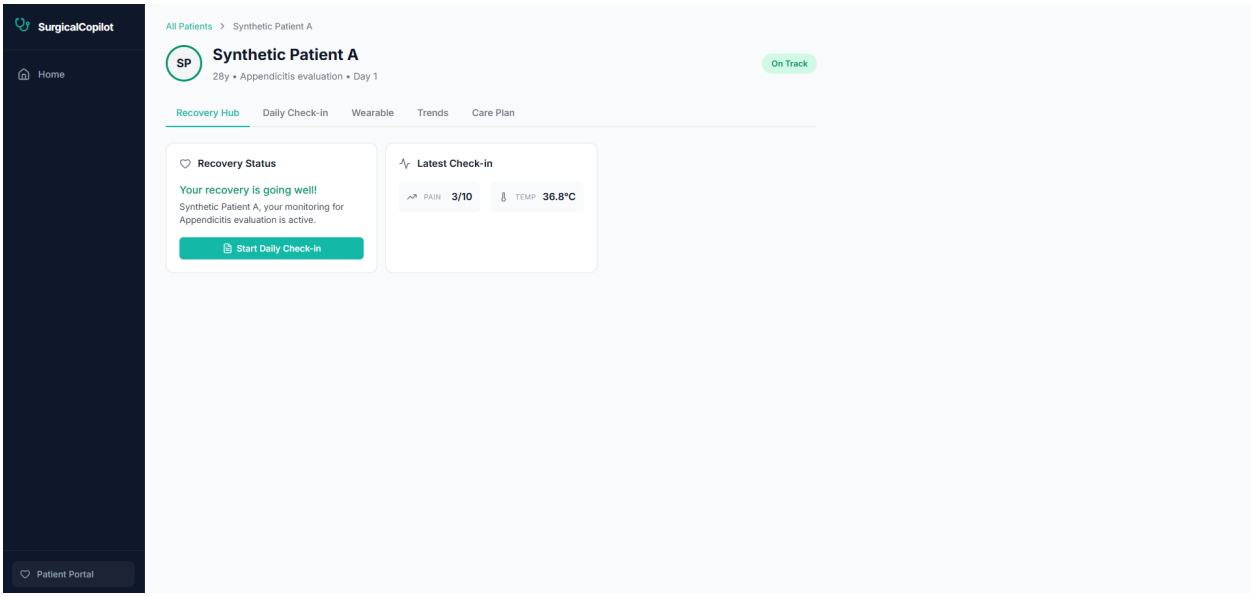
20. Voice Analysis in Progress

Live AI analysis running on voice input. Shows uploaded 22-second audio recording, "Running Analysis..." status with spinner, processing indicator showing "MedGemma 27B + Phase 1 adapter". Real-time inference feedback.



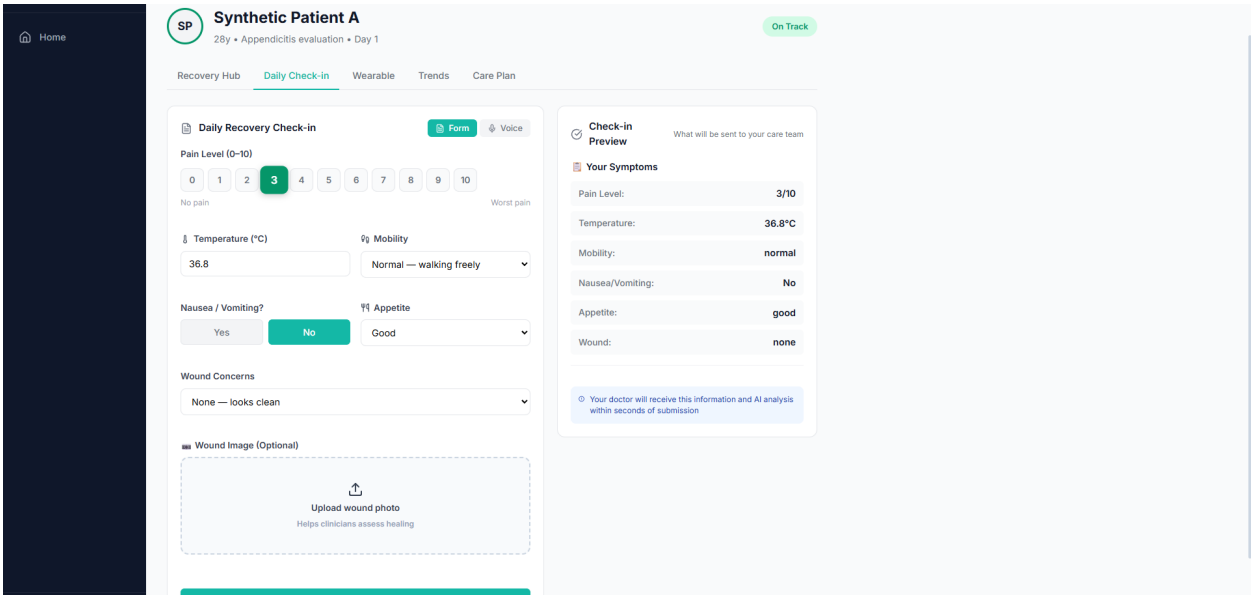
21. Patient Portal - Recovery Hub

Patient-facing Recovery Hub for Synthetic Patient A (28y, appendicitis, Day 1). Shows "On Track" status badge (green), Recovery Status message, Latest Check-in summary (pain 3/10, temp 36.8°C), and "Start Daily Check-in" CTA.



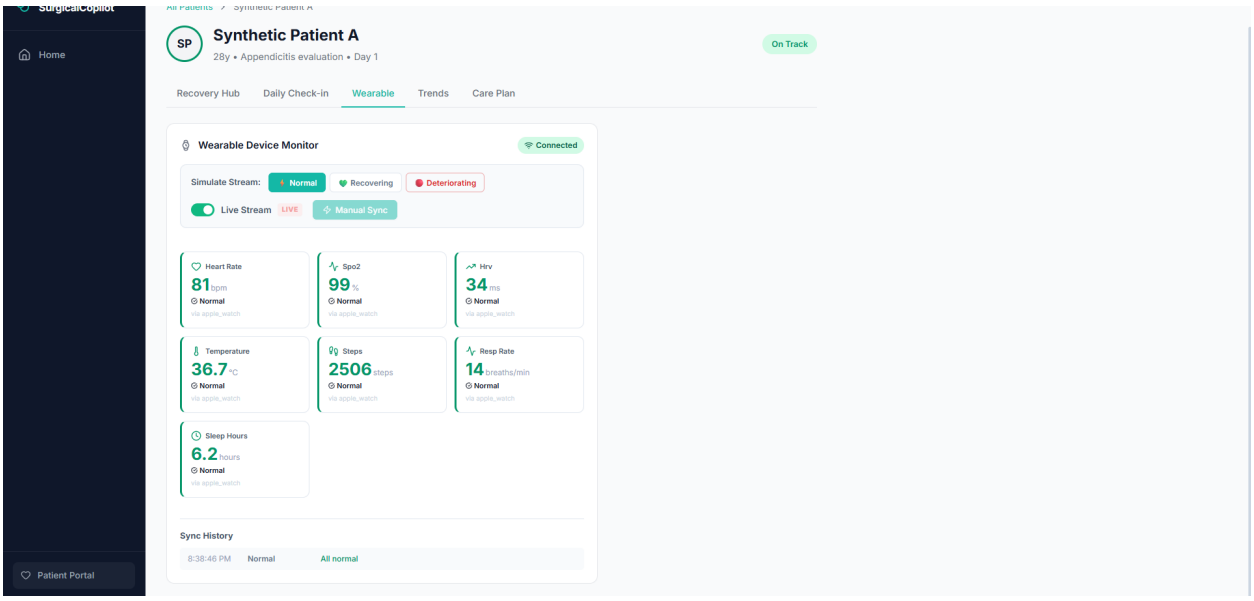
22. Patient Portal - Daily Check-in Form

Comprehensive symptom check-in form: Pain Level slider (0-10), Temperature input, Mobility dropdown, Nausea/Vomiting toggle, Appetite dropdown, Wound Concerns, and Wound Image upload. Real-time Check-in Preview shows what doctor will receive.



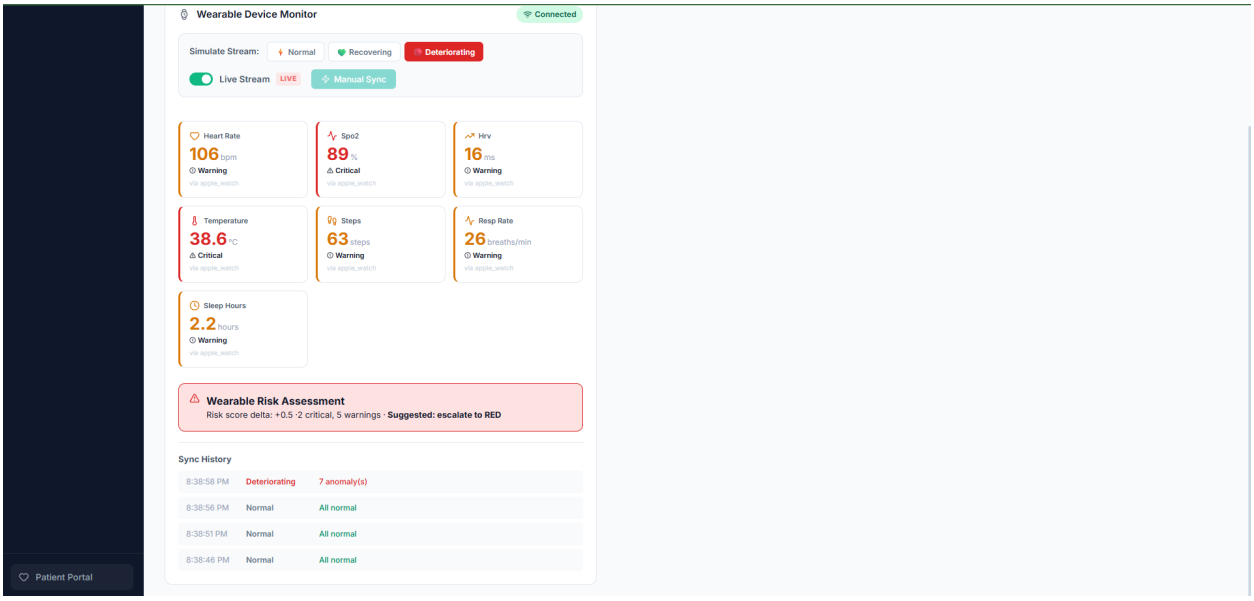
23. Wearable Monitor - Normal State

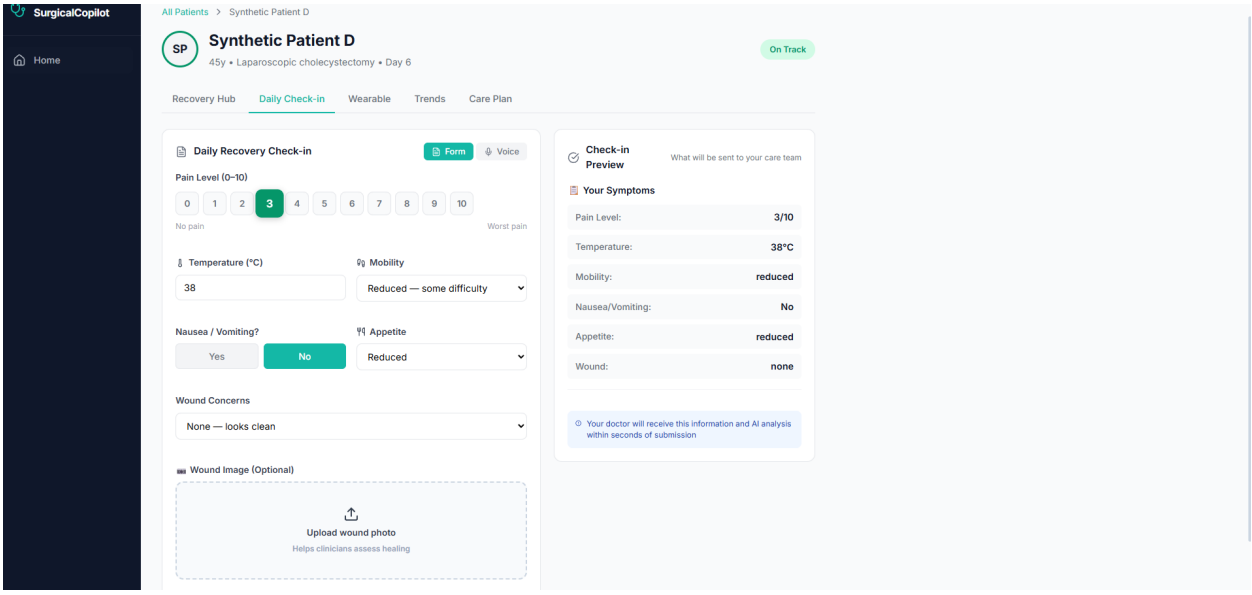
Wearable Device Monitor showing "Connected" status with Normal/Recovering/Deteriorating simulation. 7 metrics from Apple Watch: HR 81bpm, SpO2 99%, HRV 34ms, Temp 36.7°C, Steps 2506, Resp 14/min, Sleep 6.2h. All marked "Normal" (green).



24. Wearable Monitor - Deteriorating State

CRITICAL: Deteriorating patient simulation showing anomaly detection. HR 106 (Warning), SpO2 89% (Critical), HRV 16ms (Warning), Temp 38.6°C (Critical), Steps 63, Resp 26/min, Sleep 2.2h. Risk Assessment: "2 critical, 5 warnings - Suggested: escalate to RED".



25. Check-in with Warning Signs

Patient D (45y, cholecystectomy, Day 6) check-in showing concerning symptoms: Temperature 38°C (elevated), Mobility "reduced - some difficulty", Appetite "reduced". Demonstrates how system captures early deterioration signals for AI analysis.